```
import spacy from sklearn . datasets import
fetch_2enewsgroups
# Load English model n1p - Spacy . load(
"en_core_web_sm")
# Load dataset data = •train' ) texts =    # sample 5 docs (remove slice for full dataset)
# Preprocessing function
def preprocess(text) :
    doc — nlp(text.lower())

    tokens = token . lemma for token in doc if not token .is_stop
        # remove stopwords and not token. is_punct # remove
        punctuation and not token. is space # remove spaces
        and token. is_alpha # keep Only alphabets

    return tokens

# Apply preprocessing for i, text in enumerate(texts)
: cleaned = preprocess(text) print(f"\nDocument
{i+l} : " print( cleaned[ : 30] ) # show first 30 tokens
```

```
Document 1
[ 'thing' , ' subject' , 'car', 'nntp', •post', ' host', 'organization •   • university', •maryland' ,  ' college', •park', 'line',
Document 2: ['guy' , ' kuo' , 'subject' , ' clock' , 'poll' , 'final' , ' summary' , 'final' , ' i , 'clock'
, •report' , keyword' , '
Document 3: ['thomas', 'e' , 'willis ' , ' subject' , pb 'question' , ' organization ' , 'purdue' , ' university' , 'engineering' , ' computer
'
Document 4: [ 'joe' , ' green' , ' subject' , 'weitek% ' organization ' 'harris' , ' computer' , ' system' , 'division ' ' line ' , 'distribution'
,
Document 5:
[ ' jonathan', ' mcdowell ' ,    ' subject', shuttle', • launch' , 'question    ' organization • • smithsonian •   astrophysical '    obser
```

```python
import spacy
from sklearn.datasets import fetch_20newsgroups

# Load spaCy model
nlp = spacy.load("en_core_web_sm")

# Load dataset
data = fetch_20newsgroups(subset='train')
texts = data.data[:1]   # taking 1 document for clear output

# Preprocessing function
def preprocess(text):
    doc = nlp(text.lower())

    tokens = [
        token.lemma_
        for token in doc
        if not token.is_stop
        and not token.is_punct
        and not token.is_space
        and token.is_alpha
    ]

    return tokens

# Print original + processed
for text in texts:
```



```python
    print ( " ——————— ORIGINAL DOCUMENT — )
    print (text)

    print( " \n——————— ----- PREPROCESSED DOCUMENT
==========\n" )cleaned = preprocess(text) print(cleaned)
```

========== ORIGINAL DOCUMENT ==========

From: le.cxs.t@wanumdG.du (where's my thing)
Subject: WHAT car is this! ?
Nntp-Posting-Host: rac3 . wam. umd.edu
Organization: University of Maryland, College Park Lines:
15

I was wondering if anyone out there could enlighten me on this car saw the other day. It was a 2-door sports car, looked to be from the late 60s/ early 70s. It was called a Bricklin. The doors were really small. In addition, the front bumper was separate from the rest of the body. This is all I know. If anyone can tellme a model name, engine specs, years of production, where this car is made, history, or whatever info you have on this funky looking car, please e-mail.

Thanks ,

- brought to you by your neighborhood Lerxst -

========== = PREPROCESSED DOCUMENT ========== =

['thing' , ' subject' , car' [illegible] 'maryland' , college' , ' park' , 'line'